

Nabigha MOGHARBEL

BioHealth Engineering | Microbiology & Biotechnology • Data Analysis • Systems Biology

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PROFILE

BioHealth Engineering graduate with an interdisciplinary background in microbiology, bacterial biotechnology, systems biology, molecular biology and computational biology. Research experience includes bacterial growth and bioproduction modeling, genome-scale metabolic modelling, experimental work with *E.coli*, biological data analysis and machine learning. Strong scientific programming background in Python and R, complemented by previous professional software-development experience.

RESEARCH EXPERIENCE

M2 Research Intern – Systems Biology | MICROCOSME-INRIA / BIOP-LIPhy

Feb 2026 – Jul 2026 • Grenoble, France

Project: Development of coarse-grained models of bacterial growth and bioproduction

Supervision: Hidde de Jong, Delphine Ropers & Noël Scaramozzino

- Developed coarse-grained/self-replicating models of bacterial growth and bioproduction, with a focus on bacterial production of propanediol.
- Performed biological experiments with *E. coli* to generate data for model calibration and validation.
- Combined mathematical modelling, computational analysis and microbiological experiments to investigate bacterial growth and production.
- Collaborated with the Suckjoon Jun Lab at the University of California San Diego (UCSD) on single cell vs population *E.coli* experiments using fluorescent ribosome reporters. This collaboration resulted with a publication that is currently in pre-print.
- Worked in an interdisciplinary environment combining systems biology, microbiology, mathematical modelling and biotechnology.

Optional Research Intern – Systems Biology / Metabolic Modelling | MICROCOSME-INRIA

Apr 2025 – Jul 2025 • Grenoble, France

Project: Analysis of the Mycobacterium tuberculosis exo-metabolome using genome-scale metabolic modelling approaches

Supervision: Delphine Ropers

- Integrated experimental data into a genome-scale metabolic model of *M. tuberculosis*.
- Applied Flux Balance Analysis (FBA) and sampling approaches to investigate the effect of citrate on bacterial metabolism.
- Analysed model outputs to study changes in metabolic behaviour and pathways under different conditions.
- Developed experience integrating experimental biological data with computational metabolic models.

M1 Research Intern – Data Analysis & Machine Learning | EPSP, TIMC, Université Grenoble Alpes

Feb 2025 – Apr 2025 • Grenoble, France

Project: Cognitive Assessment and Symptom-Based Classification of Alzheimer's Disease Using ADNI Data

Supervision: Dominique J. Bicut

- Analysed clinical and cognitive data from the ADNI database using statistical and machine-learning approaches.
- Developed a classification framework for identifying patterns associated with Alzheimer's disease.
- Performed quantitative and qualitative data analysis and contributed to research resulting in a peer-reviewed publication.

M2 Research Intern – Molecular Biology | École Doctorale, Lebanese University

Apr 2021 – Oct 2021 • Lebanon

Project: The relationship between M-CSF1 and bone resorption in Multiple Myeloma Patients

Supervision: Dr. Iman Hajj Dib

- Conducted a literature-based investigation of M-CSF1 and its relationship with bone resorption in multiple myeloma.
- Reviewed molecular mechanisms and evaluated M-CSF as a potential target for oncological treatment.

EDUCATION

M2 BioHealth Engineering | Université Grenoble Alpes

Sep 2025 – Sep 2026 • Grenoble, France

Planned Health Graduate School Scholarship Awardee. Relevant training: radiopharmaceuticals, proteomics, machine learning & deep learning, AI for omics and infectious diseases.

M1 BioHealth Engineering | Université Grenoble Alpes

Sep 2024 – Jun 2025 • Grenoble, France

Planned Health Graduate School Scholarship Awardee. Relevant training: scientific programming with Python, statistics with Python & R, DNA systems, biotechnology of membrane and cell systems, image processing, hackathon and data challenge.

M2 Human Molecular Diagnostics | Lebanese University

Sep 2020 – Oct 2021 • Lebanon

Relevant training: molecular diagnostics, genetic counselling, cell signalling and developmental physiopathology.

M1 Cellular & Molecular Biology | Lebanese University

Sep 2019 – Jul 2020 • Lebanon

Relevant training: molecular biology, cancer biology, genetics, applied immunology, bioinformatics, toxicology, stem cells and differentiation, and cell signaling.

BSc Life & Earth Sciences | Lebanese University

Sep 2016 – Jul 2019 • Lebanon

Core training in cell biology, biochemistry, molecular biology, genetics, physiology, chemistry, ecology and botany.

PUBLICATION & PRE-PRINTS

- Fluorescent proteins misreport ribosome abundance under translation inhibition. Aniket Zodage, Justin Chen, Nabigha Mogharbel, Hidde de Jong, Suckjoon Jun. bioRxiv; doi: <https://doi.org/10.64898/2026.09.17.752513>
- Heterogeneity in diagnostic trajectories of ADNI participants. (2026). Chanidapa Winalai, Nabigha Mogharbel, Dominique J. Bicut. Journal of Neuroscience Methods (2026); doi: <https://doi.org/10.1016/j.jneumeth.2026.110748>

TECHNICAL SKILLS

Microbiology & molecular biology: Bacterial inoculation and isolation; E. coli experimental work; microplate-based assays; carbon-source sampling and quantification using enzymatic kits; PCR; SDS-PAGE; BSL-1 laboratory work.

Systems & computational biology: Genome-scale metabolic modeling; Flux Balance Analysis (FBA); metabolic network analysis; model calibration and validation; mathematical modelling and simulation; biological data integration.

Data science: Python; R; MATLAB; statistical analysis; machine learning; biological data analysis.

Computing: Git; Linux; SQL/NoSQL; JavaScript; Swift; Node.js/Express; PHP/Laravel.

ADDITIONAL PROFESSIONAL EXPERIENCE

Robotics and Programming Trainer | CodeBrave

Jun 2023 – Aug 2024 • Remote

- Taught 12 high school students programming in Python and Front-End web development enrolled in Robotics and Programming bootcamp - 4th cohort.
- Developed course materials (presentations, exercises and challenges) and assessment criteria for milestone projects.

iOS Developer | Vianeos

Aug 2023 – Aug 2024 • Tripoli, Lebanon

- Developed the Orbit Now video-on-demand and live-TV application using Swift.
- Collaborated with product, project-management, design and QA teams and contributed to application performance optimization.

Full-Stack Developer | Codi-Tech

Oct 2022 – Jun 2023 • Tripoli, Lebanon

- Developed web applications using Node.js/Express, PHP/Laravel, JavaScript, React and SQL.
- Worked with Linux/LAMP environments and Agile project-management practices.

LANGUAGES

English: Advanced.

French: B1.

Arabic: Native speaker

REFERENCES

Hidde de Jong — MICROCOSME, Inria, Grenoble, France | hidde.de-jong@inria.fr

Dominique J. Bicut — EPSP, TIMC, Grenoble, France | bicut@ill.fr